Chapter 3: Gamma, Beta, Zeta

Special Functions of Mathematical Physics: A Tourist's Guidebook

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Outline

Where we are going

The previous chapter showed that real functions are slices of complex functions. Here we extend *discrete* functions to continuous (and complex) ones, and study the named functions that result.

- **Harmonic numbers** $H_n = \sum_{k=1}^{n} \frac{1}{k}$ and their continuous interpolation; the Euler–Mascheroni constant γ.
- The **gamma function** $\Gamma(z)$: the analytic extension of the factorial. Recursion, reflection, Euler's limit, Stirling, fractional calculus, digamma.
- The **beta function** $B(p, q)$: a gamma-ratio closely related to binomial coefficients.
- The **Riemann zeta function** $\zeta(s)$: the analytic extension of the p-series, its integral and Hankel-contour forms, the values $\zeta(-n)$, and the Euler product over primes.

Detour: extending the mathematical universe

The book pauses here for the philosophical question of whether mathematical objects are invented or discovered. Either way, much of this chapter repeats a familiar working move: when an operation asks for an object outside the current universe, we enlarge the universe in the smallest useful way.

$$\mathbb{N} \xrightarrow{\text{subtraction}} \mathbb{Z} \xrightarrow{\text{division}} \mathbb{Q} \xrightarrow{\text{polynomial roots}} \overline{\mathbb{Q}} \xrightarrow{\text{limits, series}} \mathbb{R} \subset \mathbb{C}.$$

The algebraic numbers $\overline{\mathbb{Q}}$ contain roots of polynomial equations, such as $\sqrt{2}$ and the roots of x^2+1. They still miss transcendental numbers such as e and π, which arise naturally from limits, infinite series, and analytic functions.
The same completion idea applies to functions. A discrete object such as

$$H_n, \qquad n!, \qquad \sum_{k=1}^{\infty} k^{-p}$$

may be only a shadow of a better continuous or complex-analytic object. The chapter follows those shadows to $f(s)$, $\Gamma(z)$, $B(p,q)$, and $\zeta(s)$.

Table of contents

Block stacking and the harmonic numbers

How far can n identical blocks (length two units, unit mass) overhang a table edge? With one block we balance it half off, so its center of mass sits at the edge $x = 0$; the overhang is $H_1 = 1$.

Suppose $n-1$ stacked blocks already have their center of mass at the edge, with total overhang H_{n-1}. Slip one more block underneath with its center at -1, so its end lines up with the table edge. Each block has unit mass, so the new center of mass x of all n blocks is the mass-weighted average (total mass n times x equals the sum of mass times position): the top $n-1$ blocks contribute mass $n-1$ at position 0, and the new block mass 1 at position -1, so

$$n\,x = (n-1)\cdot 0 + 1\cdot(-1) \implies x = -\tfrac{1}{n}.$$

Shifting the whole stack forward by $1/n$ puts the new center of mass back at the edge. The overhang grows to $H_n = H_{n-1} + \frac{1}{n}$. With the empty-sum base case $H_0 = 0$, induction gives

$$H_n = 1 + \tfrac{1}{2} + \tfrac{1}{3} + \tfrac{1}{4} + \cdots + \tfrac{1}{n} = \sum_{k=1}^{n} \frac{1}{k}. \tag{3.1.1}$$

Block stacking and the harmonic numbers (cont.)

The series $1 + \frac{1}{2} + \frac{1}{3} + \cdots$ is the *harmonic series*; its partial sums (3.1.1) are the *harmonic numbers* $\{1, \frac{3}{2}, \frac{11}{6}, \frac{25}{12}, \ldots\}$. They diverge, but extraordinarily slowly: one needs about 10^{43} terms before $H_n > 100$.

This raises two questions we now answer.

- What *continuous* function interpolates the discrete sequence H_n?
- How slowly does the sequence diverge?

The harmonic series is itself the special case $p = 1$ of the p-series (hyperharmonic series) $\sum_{k=1}^{\infty} k^{-p}$, which returns at the end of the chapter as the zeta function.

Interpolating the partial sums

Euler's idea (1700s): find f with $f(n) = H_n = \sum_{k=1}^{n} \frac{1}{k}$. For a positive integer n the finite geometric sum gives

$$\frac{1 - x^n}{1 - x} = 1 + x + x^2 + \cdots + x^{n-1}.$$

Integrating term by term over $[0, 1]$,

$$\begin{aligned} \int_0^1 \frac{1 - x^n}{1 - x}\, dx &= \int_0^1 \left(1 + x + x^2 + \cdots + x^{n-1}\right) dx \\ &= \left(x + \tfrac{1}{2}x^2 + \tfrac{1}{3}x^3 + \cdots + \tfrac{1}{n}x^n\right)\Big|_0^1 \\ &= 1 + \tfrac{1}{2} + \tfrac{1}{3} + \cdots + \tfrac{1}{n} = H_n. \end{aligned}$$

Replacing the integer n by a continuous parameter s (initially real $s > -1$, or complex s with $x^s = e^{s \log x}$ and $\operatorname{Re} s > -1$), the function

$$f(s) = \int_0^1 \frac{1 - x^s}{1 - x}\, dx \tag{3.1.2}$$

interpolates the partial sums of the harmonic series.

Interpolating the partial sums (cont.)

For the rate of divergence, compare the sum with an integral:

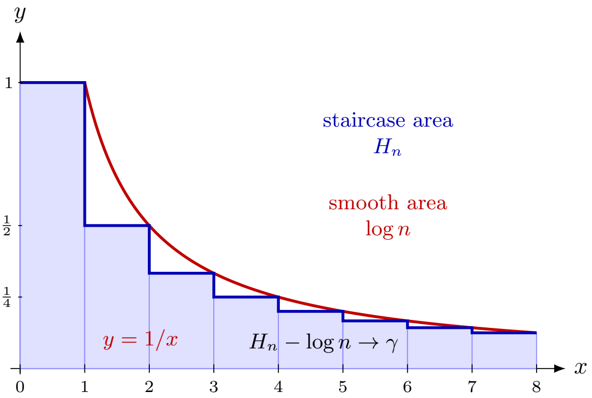
$$\sum_{k=1}^{n} \frac{1}{k} \;\approx\; \int_1^n \frac{1}{x}\,dx = \log n. \tag{3.1.3}$$

Both $\int_1^\infty \frac{1}{x}\,dx$ and $\sum_{n=1}^\infty \frac{1}{n}$ diverge, yet their *difference* converges. That limiting difference defines the *Euler–Mascheroni constant*

$$\gamma = \lim_{n\to\infty}\left[\sum_{k=1}^{n}\frac{1}{k} - \int_1^n \frac{1}{x}\,dx\right] = \lim_{n\to\infty}\left[\sum_{k=1}^{n}\frac{1}{k} - \log n\right]. \tag{3.1.4}$$

Geometrically γ is the area of the slivers between the staircase $1/k$ and the curve $1/x$. Since $1/x$ is convex and almost evenly bisects each rectangle, γ is a little above 0.5; the actual value is $\gamma = 0.5772156649\ldots$

Euler–Mascheroni as a staircase error



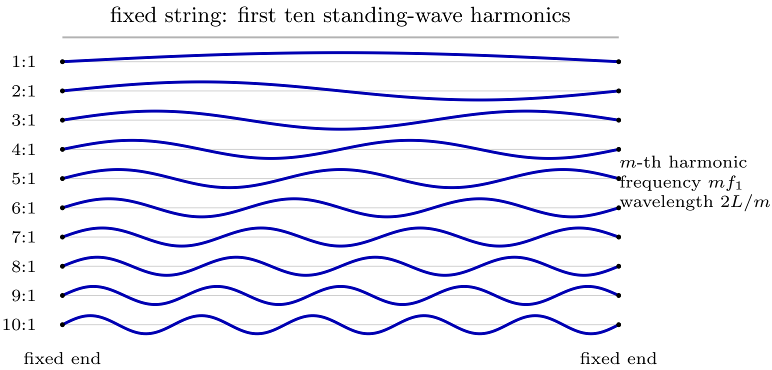
The harmonic number H_n is the area of a unit-width staircase. The logarithm $\log n = \int_1^n dx/x$ is the area under the smooth curve. The limiting excess area is $\gamma = \lim_{n\to\infty}\big(H_n - \log n\big)$.

Detour: harmonics in music

The harmonic series gets its name from string harmonics. A string fixed at both ends has normal modes

$$u_m(x) = \sin\left(\frac{m\pi x}{L}\right), \qquad m = 1, 2, 3, \ldots,$$

so the m-th mode has wavelength $2L/m$ and frequency m times the fundamental.



Detour: harmonics in music (cont.)

Low integer ratios sound consonant because their wave patterns share short common periods:

note	C	D	E	F	G	A	B	C
ratio	1:1	9:8	5:4	4:3	3:2	5:3	15:8	2:1

For example, the perfect fifth $G : C = 3 : 2$ has a small common period, while the tritone

$$F : B = (4{:}3) : (15{:}8) = 32 : 45$$

has a much larger common period and sounds more dissonant. This is a physical instance of the same arithmetic structure that appears in H_n and later in $\zeta(s)$.

Detour: harmonics in music (cont.)

Starting from a piano's C_2, the integer harmonics give C_3 at 2:1, G_3 at 3:1, C_4 at 4:1, and E_4 at 5:1. Dropping those notes by octaves produces the just-intonation ratios in the table above, such as $G = 3{:}2$ and $E = 5{:}4$.
The seventh harmonic is awkward: its ratios with the fifth and sixth harmonics have large least common multiples, so it contributes audible roughness. Piano hammers are commonly placed near one seventh of the string length to suppress that overtone.
Modern equal temperament chooses the fixed semitone ratio

$$2^{1/12},$$

which gives convenient transposition at the cost of replacing the exact just ratios by nearby approximations.

Table of contents

Definition and the factorial recursion

Just as Euler asked for a continuous extension of H_n, we ask: what continuous function extends the factorial $n!$? The answer is the *gamma function*, defined by the *Euler integral of the second kind*

$$\Gamma(z) = \int_0^\infty e^{-t}\, t^{z-1}\, dt, \qquad \mathrm{Re}(z) > 0. \tag{3.2.1}$$

Integrate $\Gamma(z+1)$ by parts, with $u = t^z$ and $dv = e^{-t}dt$ (so $du = z\, t^{z-1}dt$, $v = -e^{-t}$):

$$\begin{aligned}\Gamma(z+1) = \int_0^\infty e^{-t}\, t^z\, dt = -t^z\, e^{-t}\Big|_0^\infty + z \int_0^\infty e^{-t}\, t^{z-1}\, dt \\ = z\,\Gamma(z).\end{aligned}$$

The boundary term vanishes at both ends (for $\mathrm{Re}(z) > 0$). This is the *recursion*

$$\Gamma(z+1) = z\,\Gamma(z). \tag{3.2.2}$$

From (3.2.1), $\Gamma(1) = \int_0^\infty e^{-t}\, dt = 1$.

Definition and the factorial recursion (cont.)

Iterating the recursion from $\Gamma(1) = 1$,

$$\Gamma(2) = 1, \quad \Gamma(3) = 2 \cdot 1, \quad \Gamma(4) = 3 \cdot 2 \cdot 1, \quad \Gamma(5) = 4 \cdot 3 \cdot 2 \cdot 1, \ \ldots$$

so for nonnegative integers n,

$$\Gamma(n+1) = n!, \qquad n = 0, 1, 2, \ldots. \tag{3.2.3}$$

Note the shift: $\Gamma(n+1) = n!$, not $\Gamma(n) = n!$. (The sibling *pi function* $\Pi(z) = \Gamma(z+1)$ matches the factorial without the shift, $\Pi(n) = n!$.)
A second form, obtained by the substitution $t = s^2$ (so $dt = 2s\,ds$), is convenient at half-integers:

$$\Gamma(z) = 2\int_0^\infty e^{-s^2}\, s^{2z-1}\, ds, \qquad \operatorname{Re}(z) > 0. \tag{3.2.4}$$

Worked examples: half-integer values

Using the Gaussian form (3.2.4) with $z = \frac{1}{2}$ (so $2z - 1 = 0$),

$$\Gamma\left(\tfrac{1}{2}\right) = 2\int_0^{\infty} e^{-s^2}\,ds = \int_{-\infty}^{\infty} e^{-s^2}\,ds = \sqrt{\pi}. \tag{3.2.5}$$

The middle step uses evenness of e^{-s^2} to double the half-line into the full line, where the Gaussian integral equals $\sqrt{\pi}$.
Then the recursion (3.2.2) gives the next half-integer,

$$\Gamma\left(\tfrac{3}{2}\right) = \tfrac{1}{2}\,\Gamma\left(\tfrac{1}{2}\right) = \tfrac{1}{2}\sqrt{\pi}. \tag{3.2.6}$$

Analytic continuation to the left half-plane

The integral (3.2.1) converges only for $\mathrm{Re}(z) > 0$. To reach $\mathrm{Re}(z) \leq 0$ we run the recursion *backward*: from $\Gamma(z+1) = z\,\Gamma(z)$,

$$\Gamma(z) = \frac{1}{z}\,\Gamma(z+1) = \frac{1}{z}\int_0^\infty e^{-t}\,t^z\,dt, \qquad \mathrm{Re}(z) > -1,\ z \neq 0.$$

The right side is defined for $\mathrm{Re}(z) > -1$ except at $z = 0$, where the factor $1/z$ exposes a *simple pole*. Continuing,

$$\Gamma(z+2) = (z+1)\,\Gamma(z+1) = (z+1)z\,\Gamma(z),$$

and for arbitrary m,

$$\Gamma(z+m) = (z+m-1)\cdots(z+1)\,z\,\Gamma(z).$$

Choose the positive integer m large enough that $\mathrm{Re}(z+m) > 0$, so that $\Gamma(z+m)$ is still given by the original integral definition. Solving for $\Gamma(z)$,

$$\Gamma(z) = \frac{\Gamma(z+m)}{(z+m-1)\cdots(z+1)\,z}, \tag{3.2.7}$$

valid except where z is a non-positive integer, at which Γ has a simple pole.

Analytic continuation to the left half-plane (cont.)

Using (3.2.7) in the left half-plane is cumbersome. A cleaner tool is the reflection formula.

Theorem (Euler's reflection formula)

$$\Gamma(z)\,\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}.$$

By analytic continuation it suffices to prove this for $z = a$ with $0 < \mathrm{Re}(a) < 1$. Use the definition (3.2.1) for each factor; for the second, the exponent is $(1-a)-1 = -a$, so $\Gamma(1-a) = \int_0^\infty s^{-a} e^{-s}\, ds$. Combining the two integrals,

$$\Gamma(a)\,\Gamma(1-a) = \int_0^\infty s^{-a} e^{-s}\, ds \int_0^\infty t^{a-1} e^{-t}\, dt = \int_0^\infty \int_0^\infty s^{-a} t^{a-1}\, e^{-(s+t)}\, dt\, ds.$$

Regroup the integrand to prepare a substitution: $s^{-a} t^{a-1} = (t/s)^{a-1}\, s^{-1}$ and $e^{-(s+t)} = e^{-s(1+t/s)}$, so

$$\Gamma(a)\,\Gamma(1-a) = \int_0^\infty \int_0^\infty (t/s)^{a-1}\, e^{-s(1+t/s)}\, \frac{dt\, ds}{s}.$$

Analytic continuation to the left half-plane (cont.)

Substitute $t = su$ (so $dt = s\,du$):

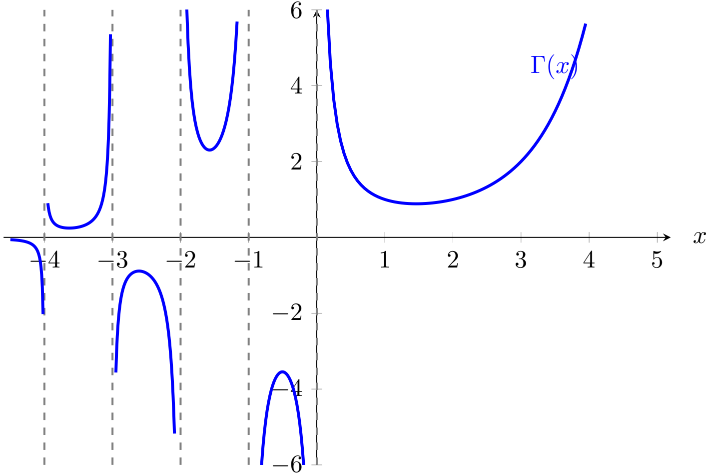
$$\Gamma(a)\,\Gamma(1-a) = \int_0^\infty \int_0^\infty u^{a-1}\, e^{-s(1+u)}\, ds\, du = \int_0^\infty u^{a-1} \left(\int_0^\infty e^{-s(1+u)}\, ds \right) du$$
$$= \int_0^\infty u^{a-1} \left(\frac{-1}{1+u}\, e^{-s(1+u)} \Big|_{s=0}^{\infty} \right) du = \int_0^\infty \frac{u^{a-1}}{1+u}\, du = \frac{\pi}{\sin(\pi a)}.$$

The last integral is the same keyhole-contour integral from Chapter 2: use the branch $0 < \arg u < 2\pi$ around the positive real-axis cut, with counterclockwise keyhole orientation. For $0 < \mathrm{Re}(a) < 1$ it gives $\int_0^\infty u^{a-1}/(1+u)\, du = \pi/\sin(\pi a)$.
To get a ready expression for negative arguments, put $w = -z$ in the reflection formula $\Gamma(w)\,\Gamma(1-w) = \pi/\sin(\pi w)$, which gives $\Gamma(-z)\,\Gamma(1+z) = \pi/\sin(-\pi z)$. Since $\sin(-\pi z) = -\sin(\pi z)$ and $\Gamma(1+z) = \Gamma(z+1)$, solving for $\Gamma(-z)$ yields

$$\Gamma(-z) = -\frac{\pi}{\Gamma(z+1)\,\sin(\pi z)}. \tag{3.2.8}$$

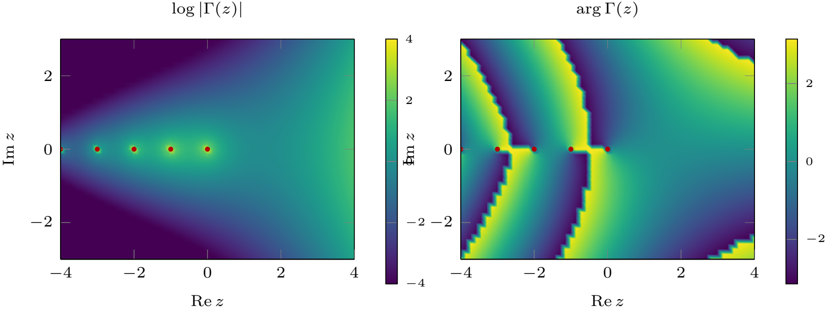
The poles of Γ at $z = 0, -1, -2, \ldots$ are visible here.

Real-axis picture of Γ



On the positive real axis, $\Gamma(n+1) = n!$. Running the recursion backward reveals simple poles at $0, -1, -2, \ldots,$ and the signs alternate between consecutive poles.

Complex-plane picture of Γ



The red dots mark the simple poles at $0, -1, -2, \ldots$. The magnitude panel shows the pole ridges and decay away from the negative real axis; the phase panel shows the winding of $\Gamma(z)$ around those poles.

Euler's limit representation

Define the truncated integral

$$F(z,n) = \int_0^n \left(1 - \frac{t}{n}\right)^n t^{z-1}\,dt, \qquad \operatorname{Re} z > 0. \tag{3.2.9}$$

As $n \to \infty$ the bracket tends to e^{-t}. Expanding the logarithm $\log(1-x) = -x - \frac{1}{2}x^2 - \cdots$ with $x = t/n$,

$$\left(1 - \frac{t}{n}\right)^n = e^{\,n\log(1-t/n)} = e^{\,n\left(-\frac{t}{n} - \frac{t^2}{2n^2} - \cdots\right)} \;\longrightarrow\; e^{-t},$$

Extend the integrand by 0 for $t > n$. For fixed $\operatorname{Re} z > 0$, $(1 - t/n)^n \le e^{-t}$ on $0 \le t \le n$, so dominated convergence justifies passing the limit through the integral:

$$\Gamma(z) = \int_0^\infty e^{-t} t^{z-1}\,dt = \lim_{n\to\infty} F(z,n).$$

Substitute $t = nu$ (so $dt = n\,du$) in (3.2.9):

$$F(z,n) = n^z \int_0^1 (1-u)^n\, u^{z-1}\,du.$$

Euler's limit representation (cont.)

Integrate by parts repeatedly. Each step trades a power of $(1-u)$ for a factor and raises the power of u:

$$
\begin{aligned}
F(z,n) &= n^z \left(\frac{n}{z} \int_0^1 (1-u)^{n-1} u^z \, du \right) \\
&= n^z \left(\frac{n(n-1)}{z(z+1)} \int_0^1 (1-u)^{n-2} u^{z+1} \, du \right) = \cdots \\
&= n^z \left(\frac{n(n-1)\cdots 1}{z(z+1)\cdots(z+n-1)} \int_0^1 u^{z+n-1} \, du \right).
\end{aligned}
$$

After n integrations by parts the power of $(1-u)$ is exhausted; the leftover integral is $\int_0^1 u^{z+n-1}\,du = 1/(z+n)$. With $n(n-1)\cdots 1 = n!$ this supplies the last denominator factor,

$$
F(z,n) = n^z \frac{n!}{z(z+1)\cdots(z+n-1)} \cdot \frac{1}{z+n} = n^z \frac{n!}{z(z+1)\cdots(z+n)}.
$$

Euler's limit representation (cont.)

Taking the limit yields *Euler's product/limit form*

$$\Gamma(z) = \lim_{n\to\infty} \frac{n^z\, n!}{z(z+1)\cdots(z+n)}, \qquad z \notin \{0, -1, -2, \dots\}. \tag{3.2.10}$$

The derivation above starts in $\operatorname{Re} z > 0$; the limit formula then extends meromorphically to the stated domain.

Summary of gamma formulas

The load-bearing identities, collected:

$$\begin{aligned}
\Gamma(z) &= \int_0^\infty e^{-t}\, t^{z-1}\, dt, \qquad \operatorname{Re}(z) > 0, \\
\Gamma(z+1) &= z\,\Gamma(z), \\
\Gamma\left(\tfrac{1}{2}\right) &= \sqrt{\pi}, \\
\Gamma(z)\,\Gamma(1-z) &= \frac{\pi}{\sin(\pi z)}, \\
\Gamma(z) &= \lim_{n\to\infty} \frac{n^z\, n!}{z(z+1)\cdots(z+n)}, \qquad z \notin \{0, -1, -2, \dots\}.
\end{aligned}$$

For integers $n \geq 0$, $\Gamma(n+1) = n!$, so $\Gamma(z)$ analytically continues the shifted factorial notation $(z-1)!$. We use these identities freely below for the beta and zeta functions.

Worked examples: gamma integrals

Example. Compute $\displaystyle\int_0^\infty e^{-x^p}\,dx$ for $p > 0$. Set $t = x^p$, so $x = t^{1/p}$ and $dx = \frac{1}{p} t^{1/p-1}\,dt$. Then

$$\int_0^\infty e^{-x^p}\,dx = \frac{1}{p}\int_0^\infty e^{-t}\,t^{(1/p)-1}\,dt = \frac{1}{p}\Gamma\left(\frac{1}{p}\right) = \Gamma\left(1+\frac{1}{p}\right). \tag{3.2.11}$$

The last step uses $\frac{1}{p}\Gamma(\frac{1}{p}) = \Gamma(1+\frac{1}{p})$ from the recursion.

Worked examples: gamma integrals (cont.)

Example (oscillatory version). Compute $\displaystyle\int_0^\infty \cos(x^p)\,dx$ for $p>1$. (For $p\le 1$ it diverges; at $p=1$ it is $\int_0^\infty \cos x\,dx$.) We find the real part of $\int_0^\infty e^{iz^p}\,dz$ using a wedge of opening angle $\pi/(2p)$. For noninteger p, choose the branch $z^p = r^p e^{ip\theta}$ in the sector $0<\theta<\pi/(2p)$ and indent the origin by a small circle of radius ε. On the punctured sector the integrand is analytic, so Cauchy's theorem gives

$$\int_\varepsilon^R e^{ix^p}\,dx + \int_{C_R} e^{iz^p}\,dz - \int_\varepsilon^R e^{i(re^{i\pi/2p})^p} e^{i\pi/2p}\,dr + \int_{C_\varepsilon} e^{iz^p}\,dz = 0.$$

The small circle contributes $O(\varepsilon)\to 0$, so the original unindented formula follows as $\varepsilon\to 0$. On the arc $z=Re^{i\theta}$, $0\le\theta\le\pi/(2p)$, the exponent has real part

$$\mathrm{Re}\big(iR^p e^{ip\theta}\big) = -R^p\sin(p\theta).$$

Using $\sin(p\theta)\ge 2p\theta/\pi$ on this interval,

$$\left|\int_{C_R} e^{iz^p}\,dz\right| \le R\int_0^{\pi/(2p)} e^{-R^p\sin(p\theta)}\,d\theta \le R\int_0^\infty e^{-(2p/\pi)R^p\theta}\,d\theta = O(R^{1-p}) \to 0$$

Worked examples: gamma integrals (cont.)

because $p > 1$. Letting $R \to \infty$ gives

$$\int_0^\infty e^{ix^p}\,dx = \lim_{R\to\infty} \int_0^{Re^{i\pi/2p}} e^{iz^p}\,dz.$$

Worked examples: gamma integrals (cont.)

On the rotated ray set $z = r\,e^{i\pi/2p}$ (so $r = z\,e^{-i\pi/2p}$). Then

$$z^p = e^{i\pi/2}\,r^p = i\,r^p, \qquad dz = dr\,e^{i\pi/2p},$$

and therefore

$$\int_0^{Re^{i\pi/2p}} e^{iz^p}\,dz = \int_0^R e^{\,i\cdot ir^p}\,e^{i\pi/2p}\,dr = e^{i\pi/2p}\int_0^R e^{-r^p}\,dr.$$

Letting $R \to \infty$ and using (3.2.11),

$$\int_0^\infty e^{iz^p}\,dz = e^{i\pi/2p}\,\Gamma\left(1+\frac{1}{p}\right).$$

Taking the real part with $e^{i\pi/2p} = \cos\frac{\pi}{2p} + i\sin\frac{\pi}{2p}$,

$$\int_0^\infty \cos(x^p)\,dx = \cos\left(\frac{\pi}{2p}\right)\Gamma\left(1+\frac{1}{p}\right). \tag{3.2.12}$$

Stirling's approximation

For large n we need a tractable form of $n!$ (already $n > 170$ overflows typical hardware). The method of steepest descent gives, for a saddle point z_0 of f,

$$\int_C g(z)\, e^{sf(z)}\, dz \approx \frac{\sqrt{2\pi}\, g(z_0)\, e^{sf(z_0)}\, i\, e^{-i\theta/2}}{\sqrt{|s\, f''(z_0)|}}, \qquad \theta = \arg f''(z_0). \tag{3.2.13}$$

Write the factorial as a gamma integral and substitute $t = nz$ ($dt = n\,dz$). Then $t^n = (nz)^n = n^n z^n$ and $e^{-t} = e^{-nz}$, so $t^n e^{-t}\, dt = n^{n+1}\, z^n e^{-nz}\, dz$; writing $z^n e^{-nz} = e^{\,n\log z - nz} = e^{\,n(\log z - z)}$,

$$n! = \Gamma(1+n) = \int_0^\infty t^n\, e^{-t}\, dt = n^{n+1} \int_0^\infty e^{\,n(\log z - z)}\, dz.$$

Thus $f(z) = \log z - z$, with

$$f'(z) = \frac{1}{z} - 1, \qquad f''(z) = -\frac{1}{z^2}.$$

Stirling's approximation (cont.)

The saddle is $f'(z) = 0$ at $z = 1$, where $f(1) = -1$ and $\theta = \arg f''(1) = \arg(-1) = \pi$, $|f''(1)| = 1$. Inserting into (3.2.13) with $s = n$, $g \equiv 1$, and the prefactor n^{n+1},

$$n! = \Gamma(n+1) \approx \frac{\sqrt{2\pi}\, n^{n+1}\, e^{-n}}{\sqrt{n}} = \sqrt{2\pi}\, n^{n+1/2}\, e^{-n}. \tag{3.2.14}$$

(The factor $i\, e^{-i\pi/2} = i(-i) = 1$ and $\sqrt{|n \cdot 1|} = \sqrt{n}$ collapse the steepest-descent constants.) Substituting (3.2.14) for both $n!$ and $m!$ in the binomial coefficient $\binom{n}{m} = \frac{n!}{m!\,(n-m)!}$ recovers the same large-n approximation found earlier. For large n one may also work with $\log \Gamma(n)$ directly.

Fractional calculus

The gamma function also extends derivatives to non-integer order once a convention is chosen. For the monomial x^n on $x > 0$ (Riemann–Liouville convention, with the usual real branch),

$$\frac{d}{dx}x^n = nx^{n-1}, \qquad \frac{d^2}{dx^2}x^n = n(n-1)x^{n-2}, \ \ldots, \qquad \frac{d^k}{dx^k}x^n = \frac{n!}{(n-k)!}\,x^{n-k}.$$

Replacing the factorials by gammas defines the k-th derivative for this convention:

$$\frac{d^k}{dx^k}x^n = \frac{\Gamma(n+1)}{\Gamma(n+1-k)}\,x^{n-k}. \tag{3.2.15}$$

For example, the half-derivative of x^2 ($n = 2$, $k = \frac{1}{2}$):

$$\frac{d^{1/2}}{dx^{1/2}}x^2 = \frac{\Gamma(3)}{\Gamma\left(\frac{5}{2}\right)}\,x^{3/2} = \frac{8}{3\sqrt{\pi}}\,x^{3/2}, \tag{3.2.16}$$

using $\Gamma(3) = 2$ and $\Gamma\left(\frac{5}{2}\right) = \frac{3}{4}\sqrt{\pi}$.

Fractional calculus (cont.)

In this monomial example, applying the half-derivative twice reproduces the ordinary derivative:

$$\frac{d}{dx}x^2 = \frac{d^{1/2}}{dx^{1/2}}\frac{d^{1/2}}{dx^{1/2}}x^2 = \frac{d^{1/2}}{dx^{1/2}}\left(\frac{8}{3\sqrt{\pi}}\,x^{3/2}\right) = \frac{8}{3\sqrt{\pi}}\,\frac{\Gamma\left(\frac{5}{2}\right)}{\Gamma(2)}\,x = 2x,$$

a consistency check ($\Gamma(\frac{5}{2})/\Gamma(2) = \frac{3}{4}\sqrt{\pi}$, and $\frac{8}{3\sqrt{\pi}} \cdot \frac{3}{4}\sqrt{\pi} = 2$).
For $x > 0$ and the Riemann–Liouville lower terminal 0, fractional derivatives extend to analytic functions term by term on closed intervals $\delta \le x \le M$ with $\delta > 0$, where the differentiated series converges uniformly. Noninteger orders produce powers such as x^{-p} near the lower terminal, so the expression may be singular at $x = 0$. Writing the exponential as a power series with gamma coefficients,

$$e^x = 1 + x + \tfrac{1}{2}x^2 + \tfrac{1}{3!}x^3 + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{\Gamma(n+1)},$$

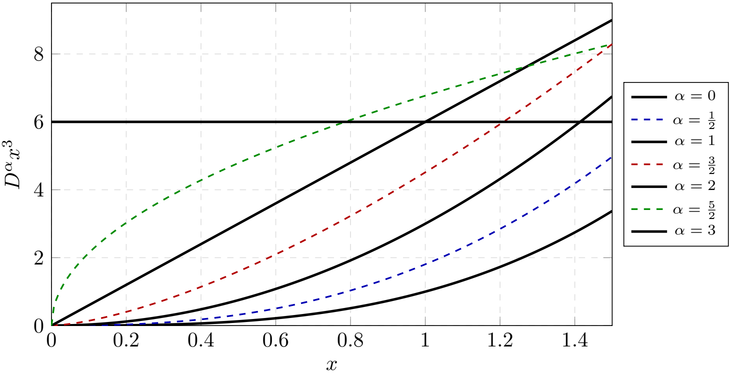
its p-th derivative is

$$\frac{d^p}{dx^p}e^x = \sum_{n=0}^{\infty} \frac{\Gamma(n+1)}{\Gamma(n+1-p)}\,\frac{1}{\Gamma(n+1)}\,x^{n-p} = \sum_{n=0}^{\infty} \frac{x^{n-p}}{\Gamma(n+1-p)}. \tag{3.2.17}$$

Fractional calculus (cont.)

Choosing $k = -1$ in (3.2.15) gives an antiderivative, $\dfrac{d^{-1}}{dx^{-1}} x^n = \dfrac{\Gamma(n+1)}{\Gamma(n+2)}\, x^{n+1}$.

Fractional derivatives of x^3



For $0 \leq \alpha \leq 3$,

$$D^\alpha x^3 = \frac{\Gamma(4)}{\Gamma(4-\alpha)} x^{3-\alpha} = \frac{6}{\Gamma(4-\alpha)} x^{3-\alpha}.$$

Integer orders recover $x^3, 3x^2, 6x, 6$. In the Riemann–Liouville convention, nearby noninteger orders past degree 3 need not vanish; they can become singular at the lower terminal $x = 0$, which is why the convention and domain matter.

The digamma function

The *logarithmic derivative* $(\log f)' = f'/f$ measures relative change. The logarithmic derivative of the gamma function is the *digamma function*

$$\psi_0(z) = \frac{d}{dz} \log \Gamma(z). \tag{3.2.18}$$

Start from Euler's limit form $\Gamma(z+1) = \lim_n \frac{n!}{(z+1)\cdots(z+n)}\, n^z$ and take the logarithm:

$$\log \Gamma(z+1) = \lim_{n\to\infty} \Big[\log n! + z \log n - \log(z+1) - \cdots - \log(z+n)\Big].$$

Differentiating term by term,

$$\begin{aligned}\psi_0(z+1) &= \lim_{n\to\infty} \left(\log n - \sum_{k=1}^{n} \frac{1}{z+k}\right)\\ &= \lim_{n\to\infty} \left(-\gamma + \sum_{k=1}^{n} \frac{1}{k} - \sum_{k=1}^{n} \frac{1}{z+k}\right) = -\gamma + \sum_{k=1}^{\infty} \frac{z}{k(k+z)},\end{aligned}$$

where $\log n = \sum_{k=1}^{n} \frac{1}{k} - \gamma + o(1)$ from (3.1.4), and the two sums combine into $\sum_k \left(\frac{1}{k} - \frac{1}{z+k}\right) = \sum_k \frac{z}{k(k+z)}$.

The digamma function (cont.)

At an integer argument $z = m$ the series telescopes against the harmonic sum:

$$\psi_0(m+1) = -\gamma + \left(\sum_{k=1}^{\infty}\frac{1}{k} - \sum_{k=1}^{\infty}\frac{1}{m+k}\right) = -\gamma + \left(\sum_{k=1}^{\infty}\frac{1}{k} - \sum_{k=m+1}^{\infty}\frac{1}{k}\right)$$
$$= -\gamma + \sum_{k=1}^{m}\frac{1}{k} = -\gamma + H_m,$$

where H_m is the m-th harmonic number. In particular $\psi_0(1) = -\gamma$ (take $m = 0$, $H_0 = 0$), which pins the constant γ to the gamma function.

Worked example: the Jeep problem

A Jeep travels one unit of distance per drum of fuel and carries at most one drum. By caching fuel along the way it pushes a base camp deeper into the desert. If j drums remain at a camp, moving the whole camp forward by a distance Δ costs j forward trips and $j-1$ return trips, hence $(2j-1)\Delta$ drums of fuel. Spending one drum at that stage gives $\Delta = 1/(2j-1)$. Summing stages $j = n, n-1, \ldots, 1$, the total distance reachable on n drums is the sum of odd-denominator reciprocals:

$$\begin{aligned}\sum_{k=1}^{n} \frac{1}{2k-1} = \sum_{k=1}^{2n-1} \frac{1}{k} - \sum_{k=1}^{n-1} \frac{1}{2k} &= H_{2n-1} - \tfrac{1}{2} H_{n-1} \\ &= \psi_0(2n) - \tfrac{1}{2}\,\psi_0(n) + \tfrac{1}{2}\gamma.\end{aligned}$$

The first line splits the full harmonic sum to $2n-1$ into its odd terms (the wanted sum) and its even terms; the even denominators are $2, 4, \ldots, 2(n-1)$, so $\sum_{k=1}^{n-1} \frac{1}{2k} = \frac{1}{2} \sum_{k=1}^{n-1} \frac{1}{k} = \frac{1}{2} H_{n-1}$. The second line uses $H_m = \psi_0(m+1) + \gamma$ (from $\psi_0(m+1) = -\gamma + H_m$), so that $H_{2n-1} - \frac{1}{2} H_{n-1} = (\psi_0(2n) + \gamma) - \frac{1}{2}(\psi_0(n) + \gamma) = \psi_0(2n) - \frac{1}{2}\psi_0(n) + \frac{1}{2}\gamma$.

Table of contents

Definition and trigonometric form

The *Euler beta function* is a gamma-ratio closely related to binomial coefficients and useful for many definite integrals. Define

$$B(p,q) = \int_0^1 t^{p-1}(1-t)^{q-1}\,dt, \qquad \operatorname{Re} p > 0, \quad \operatorname{Re} q > 0. \tag{3.3.1}$$

For nonnegative integers m, n, this relation is concrete:

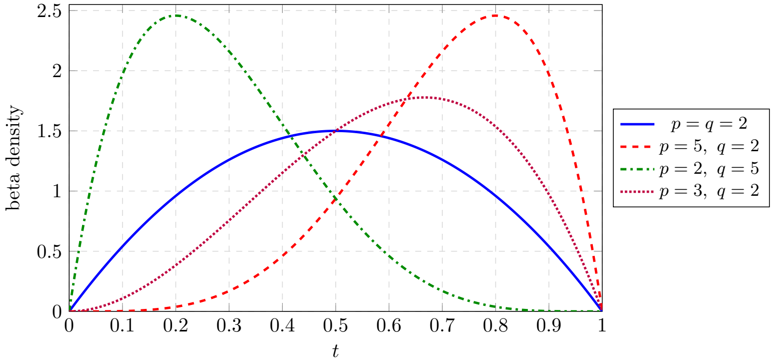
$$B(m+1, n+1) = \frac{m!\,n!}{(m+n+1)!} = \frac{1}{(m+n+1)\binom{m+n}{m}}.$$

Substitute $t = \sin^2\theta$, so $dt = 2\cos\theta\sin\theta\,d\theta$ and $1-t = \cos^2\theta$; as t runs $0 \to 1$, θ runs $0 \to \pi/2$:

$$\begin{aligned} B(p,q) &= \int_0^{\pi/2} \sin^{2p-2}\theta\,\cos^{2q-2}\theta\,(2\cos\theta\sin\theta)\,d\theta \\ &= 2\int_0^{\pi/2} \cos^{2q-1}\theta\,\sin^{2p-1}\theta\,d\theta. \end{aligned} \tag{3.3.2}$$

Here $(1-t)^{q-1} = (\cos^2\theta)^{q-1} = \cos^{2q-2}\theta$ and $t^{p-1} = \sin^{2p-2}\theta$, and the extra $2\cos\theta\sin\theta$ raises each power by one.

Beta weights on $[0, 1]$



After normalization by $B(p, q)$, the integrand $t^{p-1}(1-t)^{q-1}$ becomes a beta density. Larger p pushes weight toward 1, larger q pushes weight toward 0, and $p = q$ is symmetric.

Beta in terms of gamma

Theorem

$$B(p,q) = \frac{\Gamma(p)\,\Gamma(q)}{\Gamma(p+q)} \qquad (\operatorname{Re} p, \operatorname{Re} q > 0).$$

Write each gamma in Gaussian form (3.2.4), $\Gamma(p) = 2\int_0^\infty e^{-x^2} x^{2p-1}\,dx$, and likewise for $\Gamma(q)$ in y. Their product is a double integral over the first quadrant:

$$\Gamma(p)\,\Gamma(q) = 4\int_0^\infty\int_0^\infty e^{-(x^2+y^2)}\,x^{2p-1}\,y^{2q-1}\,dx\,dy.$$

Switch to polar coordinates $x = r\cos\theta$, $y = r\sin\theta$, with area element $r\,dr\,d\theta$ (the first quadrant is $0 \le \theta \le \pi/2$), so $x^{2p-1} = r^{2p-1}\cos^{2p-1}\theta$ and $y^{2q-1} = r^{2q-1}\sin^{2q-1}\theta$:

$$\Gamma(p)\,\Gamma(q) = 4\int_0^\infty\int_0^{\pi/2} e^{-r^2}\,r^{2p-1}\cos^{2p-1}\theta\,r^{2q-1}\sin^{2q-1}\theta\;r\,dr\,d\theta.$$

Beta in terms of gamma (cont.)

The integrand separates: collecting the r powers, $r^{2p-1}\,r^{2q-1}\,r = r^{2(p+q)-1}$, and splitting $4 = 2\cdot 2$, the double integral is a product of a radial and an angular integral,

$$\Gamma(p)\,\Gamma(q) = \underbrace{2\int_0^\infty e^{-r^2}\,r^{2(p+q)-1}\,dr}_{=\ \Gamma(p+q)}\cdot\underbrace{2\int_0^{\pi/2}\cos^{2p-1}\theta\,\sin^{2q-1}\theta\,d\theta}_{=\ B(q,p)=B(p,q)}.$$

The radial factor is $\Gamma(p+q)$ by (3.2.4). The angular factor is the trigonometric beta integral (3.3.2) with the roles of p and q swapped, hence equals $B(q,p)$; and B is symmetric, $B(q,p) = B(p,q)$ (swap $t \mapsto 1-t$ in (3.3.1)). Dividing gives the theorem. □

Summary of beta formulas

$$B(p,q) = \frac{\Gamma(p)\,\Gamma(q)}{\Gamma(p+q)},$$

$$B(m+1,n+1) = \frac{m!\,n!}{(m+n+1)!} = \frac{1}{(m+n+1)\binom{m+n}{m}},$$

$$B(p,q) = \int_0^1 t^{p-1}(1-t)^{q-1}\,dt, \quad \operatorname{Re} p, \operatorname{Re} q > 0,$$

$$B(p,q) = 2\int_0^{\pi/2} \sin^{2p-1}\theta\,\cos^{2q-1}\theta\,d\theta.$$

The first line lets us evaluate many integrals in closed form once we recognize their $t^{p-1}(1-t)^{q-1}$ shape.

Worked examples: beta integrals

Example. Compute $\int_0^1 \frac{dx}{\sqrt{1-x^4}}$. Let $u = x^4$; then $u^{1/4} = x$ and $\frac{1}{4}u^{-3/4}\,du = dx$, so

$$\int_0^1 (1-x^4)^{-1/2}\,dx = \int_0^1 (1-u)^{-1/2}\,\tfrac{1}{4}\,u^{-3/4}\,du = \tfrac{1}{4}\,B\left(\tfrac{1}{4}, \tfrac{1}{2}\right). \tag{3.3.3}$$

Matching $u^{p-1}(1-u)^{q-1}$ gives $p - 1 = -\frac{3}{4}$ $(p = \frac{1}{4})$ and $q - 1 = -\frac{1}{2}$ $(q = \frac{1}{2})$.

Worked examples: beta integrals (cont.)

Example. Compute $\displaystyle\int_0^{\pi} \sin^n x\, dx$ for $n > -1$ (in particular for every nonnegative integer n). By symmetry about $x = \pi/2$ and the trigonometric form (3.3.2), $2\int_0^{\pi/2} \cos^{2q-1}\theta\, \sin^{2p-1}\theta\, d\theta = B(p,q)$. Matching $\sin^n$ (no cosine) means $2p-1 = n$ and $2q-1 = 0$, i.e. $p = \frac{n+1}{2}$, $q = \frac{1}{2}$, giving $B\left(\frac{n+1}{2}, \frac{1}{2}\right) = B\left(\frac{1}{2}, \frac{n+1}{2}\right)$ by symmetry of B:

$$\begin{aligned}\int_0^{\pi} \sin^n x\, dx = 2\int_0^{\pi/2} \sin^n x\, dx &= B\left(\tfrac{1}{2}, \tfrac{n+1}{2}\right) \\ &= \frac{\Gamma\left(\frac{n+1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{n}{2}+1\right)} = \sqrt{\pi}\,\frac{\Gamma\left(\frac{n}{2}+\frac{1}{2}\right)}{\Gamma\left(\frac{n}{2}+1\right)}. \end{aligned} \tag{3.3.4}$$

The two factors of 2 cancel: $\int_0^{\pi/2} \sin^n = \frac{1}{2}B(\frac{1}{2}, \frac{n+1}{2})$, so $2\int_0^{\pi/2} \sin^n = B(\frac{1}{2}, \frac{n+1}{2})$, and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

Worked examples: beta integrals (cont.)

Example (volume of the n-ball). The volume $V_n(r)$ of an n-ball of radius r can be computed by slicing perpendicular to one axis. At height rx, the slice is an $(n-1)$-ball of radius $r\sqrt{1-x^2}$. Since lengths in $(n-1)$ dimensions scale volumes by the $(n-1)$st power, $V_{n-1}(r\rho) = \rho^{n-1} V_{n-1}(r)$, so

$$V_n(r) = r\, V_{n-1}(r) \int_{-1}^{1} \left(\sqrt{1-x^2}\right)^{n-1} dx.$$

The integrand is even, so $\int_{-1}^{1} = 2\int_0^1$. With $u = x^2$ on $[0,1]$, $du = 2x\,dx = 2\sqrt{u}\,dx$, so $dx = \frac{1}{2}u^{-1/2}du$; the factor 2 from evenness cancels this $\frac{1}{2}$:

$$\begin{aligned}\int_{-1}^{1}(1-x^2)^{(n-1)/2}\,dx &= 2\int_0^1(1-x^2)^{(n-1)/2}\,dx = 2\int_0^1(1-u)^{(n-1)/2}\,\tfrac{1}{2}u^{-1/2}\,du \\ &= \int_0^1(1-u)^{(n-1)/2}\,u^{-1/2}\,du = B\left(\tfrac{1}{2}, \tfrac{n+1}{2}\right),\end{aligned}$$

matching $u^{p-1}(1-u)^{q-1}$ with $p-1 = -\frac{1}{2}$ ($p=\frac{1}{2}$) and $q-1 = \frac{n-1}{2}$ ($q = \frac{n+1}{2}$). Hence $V_n(r) = r\,B\left(\frac{1}{2}, \frac{n}{2}+\frac{1}{2}\right) V_{n-1}(r)$. Writing $B = \Gamma(\frac{1}{2})\Gamma(\frac{n+1}{2})/\Gamma(\frac{n}{2}+1)$ and telescoping

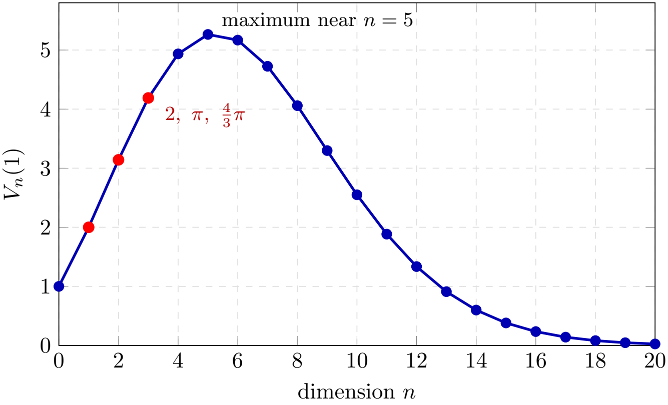
Worked examples: beta integrals (cont.)

down to $V_0(r) = 1$, the $\Gamma(\frac{n+1}{2})$ factors cancel pairwise, leaving $\Gamma(\frac{1}{2})^n = \pi^{n/2}$ in the numerator:

$$V_n(r) = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)}\, r^n. \tag{3.3.5}$$

This reproduces $V_1 = 2r$, $V_2 = \pi r^2$, $V_3 = \frac{4}{3}\pi r^3$, and extends to fractional dimension.

Volume of the unit n-ball



For radius $r = 1$ the volume

$$V_n(1) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)}$$

initially grows because there are more directions to move, then decays because most of the high-dimensional cube lies far from the unit ball.

Table of contents

From the p-series to $\zeta(s)$

Generalize the harmonic series $\sum_k k^{-1}$ to the hyperharmonic or p-series $\sum_{k=1}^{\infty} k^{-p}$. Its partial sums are the generalized harmonic numbers $H_{n,p} = \sum_{k=1}^{n} k^{-p}$, and for $p > 1$ the series converges. The *Riemann zeta function* extends p to complex values:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \qquad \mathrm{Re}(s) > 1. \tag{3.4.1}$$

For $\mathrm{Re}(s) > 1$ there is an equivalent integral representation,

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^{\infty} \frac{t^{s-1}}{e^t - 1}\, dt, \qquad \mathrm{Re}(s) > 1. \tag{3.4.2}$$

From the p-series to $\zeta(s)$ (cont.)

To prove the equivalence for $\mathrm{Re}(s) > 1$, expand $1/(e^t - 1)$ as a geometric series in e^{-t} and integrate term by term. Tonelli's theorem justifies the swap because the absolute-value series integrates to $\Gamma(\mathrm{Re}\, s) \sum_{n=1}^{\infty} n^{-\mathrm{Re}\, s} < \infty$:

$$\int_0^\infty \frac{t^{s-1}}{e^t - 1}\, dt = \int_0^\infty e^{-t}\, \frac{t^{s-1}}{1 - e^{-t}}\, dt = \int_0^\infty e^{-t}\, t^{s-1} \sum_{n=0}^{\infty} \left(e^{-t}\right)^n dt$$

$$= \int_0^\infty t^{s-1} \sum_{n=1}^{\infty} e^{-nt}\, dt = \sum_{n=1}^{\infty} \int_0^\infty t^{s-1}\, e^{-nt}\, dt.$$

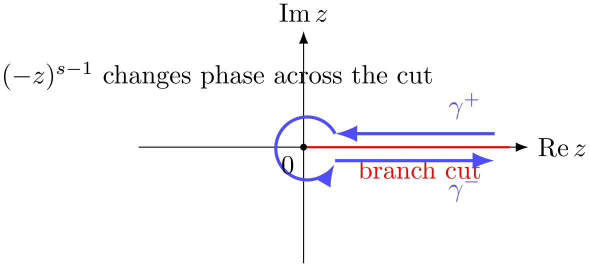
Substitute $t \mapsto t/n$ ($dt \mapsto dt/n$) in each integral:

$$= \sum_{n=1}^{\infty} n^{-s} \int_0^\infty t^{s-1}\, e^{-t}\, dt = \zeta(s)\, \Gamma(s).$$

Dividing by $\Gamma(s)$ gives (3.4.2).

Analytic continuation by a Hankel contour

To reach $\mathrm{Re}(s) < 1$, replace the half-line by a *Hankel contour* γ: in from $+\infty$ just above the positive real axis, around the origin, and back out below it. For $\mathrm{Re}(s) > 1$ define



$$\phi(s) = \int_\gamma \frac{(-z)^{s-1}}{e^z - 1}\, dz = \left(\int_R^\varepsilon + \int_{C_\varepsilon} + \int_\varepsilon^R \right) \frac{(-z)^{s-1}}{e^z - 1}\, dz. \tag{3.4.3}$$

The small circle C_ε contributes nothing as $\varepsilon \to 0$ in this $\mathrm{Re}(s) > 1$ comparison (by the *ML*-estimate), so only the two straight pieces survive as $\varepsilon \to 0,\ R \to \infty$. The factor $(-z)^{s-1} = e^{(s-1)(\log z - i\pi)}$ has a branch cut along the positive real axis: above it $\log(-z) = \log r - i\pi$, below it $\log(-z) = \log r + i\pi$, with $r = |z|$.

Analytic continuation by a Hankel contour (cont.)

Combining the upper ($\infty \to 0$) and lower ($0 \to \infty$) edges,

$$\begin{aligned}
\phi(s) &= \int_\infty^0 \frac{e^{(s-1)(\log r - i\pi)}}{e^r - 1}\, dr + \int_0^\infty \frac{e^{(s-1)(\log r + i\pi)}}{e^r - 1}\, dr \\
&= \left(-e^{-i\pi(s-1)} + e^{i\pi(s-1)}\right) \int_0^\infty \frac{e^{(s-1)\log r}}{e^r - 1}\, dr \\
&= \left(e^{-i\pi s} - e^{i\pi s}\right) \int_0^\infty \frac{r^{s-1}}{e^r - 1}\, dr = -2i\sin(\pi s) \int_0^\infty \frac{r^{s-1}}{e^r - 1}\, dr.
\end{aligned}$$

The bracket simplifies because $e^{\pm i\pi(s-1)} = -e^{\pm i\pi s}$. Using (3.4.2) for the remaining integral,

$$\phi(s) = -2i\sin(\pi s)\,\Gamma(s)\,\zeta(s) = -\,2i\pi\,\frac{\zeta(s)}{\Gamma(1-s)}, \tag{3.4.4}$$

where the second equality uses reflection $\sin(\pi s)\Gamma(s) = \pi/\Gamma(1-s)$.

The continued zeta function

Solving (3.4.4) for ζ,

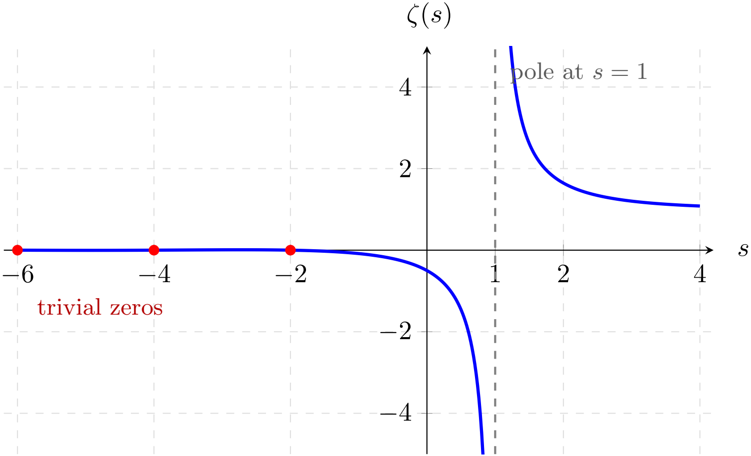
$$\zeta(s) = \frac{-1}{2i\sin(\pi s)\,\Gamma(s)}\,\phi(s) = \frac{-1}{2i\sin(\pi s)\,\Gamma(s)}\int_\gamma \frac{(-z)^{s-1}}{e^z-1}\,dz.$$

Applying Euler's reflection formula once more,

$$\zeta(s) = \frac{i\,\Gamma(1-s)}{2\pi}\,\phi(s) = \frac{i\,\Gamma(1-s)}{2\pi}\int_\gamma \frac{(-z)^{s-1}}{e^z-1}\,dz. \tag{3.4.5}$$

For $\mathrm{Re}(s) > 1$ this agrees with the original zeta integral. For continuation, keep one fixed Hankel contour with positive radius around 0; then the integral is analytic in s, and only after evaluating special values do we shrink the contour when allowed. The right-hand side gives the analytic continuation in s. The possible poles of $\Gamma(1-s)$ at $s = 1, 2, 3, \ldots$ are canceled by corresponding zeros of $\phi(s)$ except at $s = 1$, where ζ has its simple pole. The sign comes from $1/(-2i) = i/2$.

Real-axis picture of ζ



The Dirichlet series defines $\zeta(s)$ only for $s > 1$ on this real slice. Analytic continuation extends it leftward, leaving a simple pole at $s = 1$ and zeros at the negative even integers.

Worked example: $\zeta(-n)$ and Bernoulli numbers

Use the continued form (3.4.5) to evaluate $\zeta(-n)$ for $n = 0, 1, 2, \ldots$. Set $s = -n$, so the exponent in the integrand is $s - 1 = -(n+1)$ and the numerator is $(-z)^{-(n+1)}$. Pull out the sign by $(-z)^{-(n+1)} = (-1)^{-(n+1)} z^{-(n+1)} = (-1)^{n+1} z^{-(n+1)}$ (an even/odd power of -1 is unchanged by the minus sign in the exponent). The Bernoulli numbers are defined by the Laurent expansion

$$\frac{1}{e^z - 1} = \sum_{m=0}^{\infty} \frac{B_m}{m!}\, z^{m-1}, \qquad 0 < |z| < 2\pi,$$

with

$$\{B_0, B_1, B_2, \ldots\} = \{1, -\tfrac{1}{2}, \tfrac{1}{6}, 0, -\tfrac{1}{30}, 0, \tfrac{1}{42}, 0, \ldots\}.$$

Multiplying the two factors and adding exponents $-(n+1) + (m-1) = m - (n+2)$,

$$\frac{(-z)^{-(n+1)}}{e^z - 1} = (-1)^{n+1} \sum_{m=0}^{\infty} \frac{B_m}{m!}\, z^{m-(n+2)}.$$

Worked example: $\zeta(-n)$ and Bernoulli numbers (cont.)

For $s = -n$ the factor $(-z)^{-(n+1)}$ is single-valued up to the fixed sign shown above. Thus the two long sides of the Hankel contour cancel when the contour is shrunk to a small positively oriented circle around 0. The integral therefore picks out $2\pi i$ times the residue at $z = 0$, i.e. the coefficient of z^{-1}. That term has $m - (n+2) = -1$, so $m = n + 1$:

$$\int_\gamma \frac{(-z)^{-(n+1)}}{e^z - 1}\, dz = 2\pi i \ \mathrm{Res}_{z=0} = 2\pi i\, (-1)^{n+1} \frac{B_{n+1}}{(n+1)!}.$$

Worked example: $\zeta(-n)$ and Bernoulli numbers (cont.)

Insert this into (3.4.5) with $s = -n$ (so $\Gamma(1-s) = \Gamma(1+n) = n!$):

$$\zeta(-n) = \frac{i\,n!}{2\pi} \cdot 2\pi i\,(-1)^{n+1}\,\frac{B_{n+1}}{(n+1)!} = i^2(-1)^{n+1}\,\frac{B_{n+1}}{n+1}.$$

Since $i^2 = -1$, the closed form is

$$\zeta(-n) = (-1)^n\,\frac{B_{n+1}}{n+1}, \qquad n = 0, 1, 2, 3, \ldots \tag{3.4.6}$$

(using the convention $B_1 = -\frac{1}{2}$). The first values are

$$\{\zeta(0), \zeta(-1), \zeta(-2), \ldots\} = \left\{-\tfrac{1}{2},\ -\tfrac{1}{12},\ 0,\ \tfrac{1}{120},\ 0,\ -\tfrac{1}{252},\ 0,\ \tfrac{1}{240},\ 0, \ldots\right\}.$$

In particular $\zeta(0) = B_1 = -\frac{1}{2}$ and $\zeta(-1) = -B_2/2 = -\frac{1}{12}$.

Worked example: $\zeta(-n)$ and Bernoulli numbers (cont.)

This is the source of the notorious assignments

$$1+1+1+1+\cdots \overset{\zeta}{=} -\tfrac{1}{2}, \qquad 1+2+3+4+\cdots \overset{\zeta}{=} -\tfrac{1}{12}.$$

The slogan is shorthand, not summation: the Dirichlet series (3.4.1) converges only for $\mathrm{Re}(s) > 1$. The values $\zeta(0) = -\frac{1}{2}$, $\zeta(-1) = -\frac{1}{12}$ are the *analytic continuation* (3.4.5) evaluated at $s = 0, -1$, where the original series no longer applies.

The Euler product over primes

Pick $n \geq 2$ integers at random. The probability that a prime p divides any one of them is $1/p$, so all n are divisible by p with probability p^{-n}, and *not* all divisible with probability $1 - p^{-n}$. The chance that no prime divides all n integers (they share no common factor) is the product over primes $\prod_p (1 - p^{-n})$, which we will identify with $1/\zeta(n)$. For a concrete instance from the book, the four integers

$$1625, \quad 4030, \quad 6032, \quad 923$$

are all divisible by 13.

Start from the Dirichlet series

$$\zeta(n) = 1 + 2^{-n} + 3^{-n} + 4^{-n} + \cdots . \tag{3.4.7}$$

Multiply by 2^{-n} and subtract from (3.4.7); this removes every term whose index is a multiple of 2:

$$(1 - 2^{-n})\,\zeta(n) = 1 + 3^{-n} + 5^{-n} + 7^{-n} + \cdots . \tag{3.4.8}$$

The Euler product over primes (cont.)

Multiply (3.4.8) by 3^{-n} and subtract again, removing multiples of 3:

$$(1-3^{-n})(1-2^{-n})\,\zeta(n) = 1 + 5^{-n} + 7^{-n} + 11^{-n} + 13^{-n} + \cdots . \tag{3.4.9}$$

Continuing this sieve through $5^{-n}, 7^{-n}, 11^{-n}, \ldots$ strips out every composite index, leaving only 1 on the right:

$$\cdots(1-7^{-n})(1-5^{-n})(1-3^{-n})(1-2^{-n})\,\zeta(n) = 1.$$

Therefore the *Euler product*,

$$\zeta(n) = \prod_{p \text{ prime}} \frac{1}{1-p^{-n}}, \qquad n > 1. \tag{3.4.10}$$

With $\zeta(2) = \pi^2/6$ this gives about a 40% chance that two random integers share a factor ($1/\zeta(2) \approx 0.61$ that they are coprime); for four integers the chance of a common factor drops to about 8%.

Zeros, the Riemann Hypothesis, and primes

From (3.4.6), ζ vanishes at the negative even integers (the *trivial* zeros, where $B_{n+1} = 0$). All other (nontrivial) zeros are conjectured to lie on the critical line $\mathrm{Re}(s) = \frac{1}{2}$:

Riemann Hypothesis

Every nontrivial zero of $\zeta(s)$ has real part $\frac{1}{2}$.

This is one of the Clay Millennium Problems, open since Riemann (1859).
The zeta function controls the distribution of primes. The Prime Number Theorem states

$$\pi(N) \sim \frac{N}{\log N},$$

where $\pi(N)$ counts primes up to N. Through the Euler product (3.4.10), ζ links complex analysis to number theory.

Application: Zipf's law and the zeta distribution

George Zipf observed that ranked linguistic data often follow a reciprocal law. In a corpus, the frequency of a word is approximately inversely proportional to its rank:

$$f_k = \frac{k^{-1}}{H_n},$$

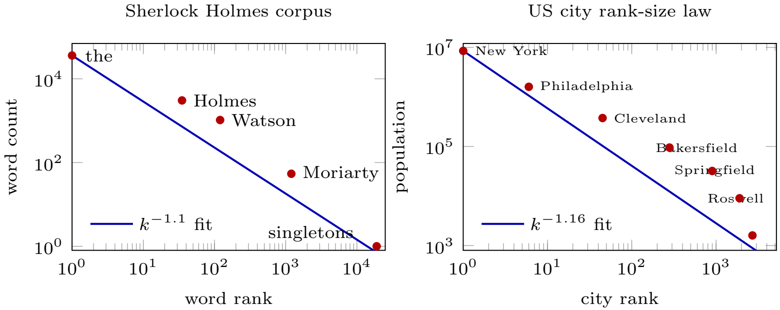
with H_n the n-th harmonic number (a normalized harmonic distribution). For English ($n \approx 170{,}000$ unique words, $H_n \approx \log n + \gamma \approx 12.62$) the top word should appear about once in thirteen words.

The *zeta distribution* sharpens this to a power law,

$$f_k = \frac{k^{-s}}{H_{n,s}}, \qquad H_{n,s} = \sum_{j=1}^{n} j^{-s}. \tag{3.4.11}$$

In the ideal infinite-rank model, replace $H_{n,s}$ by $\zeta(s)$ (for $s > 1$). Empirical rank laws often have exponents near 1. The book's examples include word ranks from large corpora, the simplified vocabulary in *Thing Explainer*, Sherlock Holmes word counts (about $s = 1.1$), and a rank-size fit to US city populations (about $s = 1.16$).

Application: Zipf's law and the zeta distribution (cont.)



Application: Zipf's law and the zeta distribution (cont.)

As a separate continuous power-law model, use the *Pareto distribution*

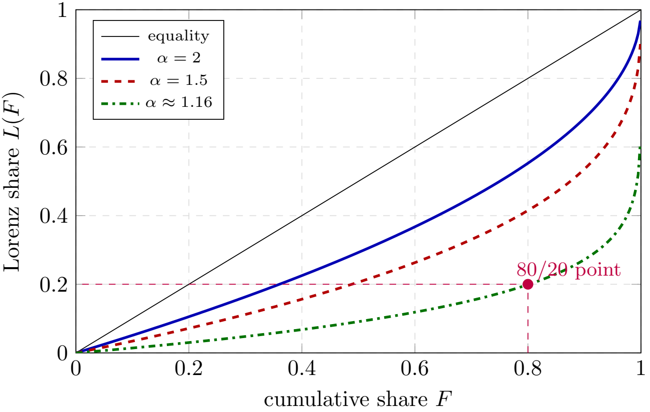
$$f(x) = \alpha\, x^{-(\alpha+1)}, \qquad x \geq 1, \quad \alpha > 0, \tag{3.4.12}$$

with cumulative distribution $F(x) = \int_1^x f(t)\,dt = 1 - x^{-\alpha}$. This is the continuous analogue of a rank law k^{-s} when the density exponent satisfies $\alpha + 1 = s$, i.e. $\alpha = s - 1$. Disparity is visualized by the *Lorenz curve*, cumulative share of an effect against cumulative share of causes. This normalization requires finite mean, so assume $\alpha > 1$. For the Pareto law, inverting F gives $x(F) = (1 - F)^{-1/\alpha}$ and the Lorenz curve

$$L(F) = 1 - (1 - F)^{1 - 1/\alpha}. \tag{3.4.13}$$

The diagonal $L = F$ is the *line of equality*. The *Gini coefficient* G (twice the area between the curve and the diagonal) is a measure of inequality: $G = 0$ is perfect equality, and for the finite-mean Pareto law $G = (2\alpha - 1)^{-1}$.

Application: Zipf's law and the zeta distribution (cont.)



Application: Zipf's law and the zeta distribution (cont.)

The *Pareto principle* (the 80/20 rule) says 80% of effects come from 20% of causes. On the Lorenz curve this means the bottom 80% of causes account for only 20% of the effects, i.e. the point $(F, L) = (0.8, 0.2)$. Solving (3.4.13) for α,

$$\alpha = \left(1 - \frac{\log(1-L)}{\log(1-F)}\right)^{-1}. \tag{3.4.14}$$

For $L = 0.2$ and $F = 0.8$ this gives $\alpha \approx 1.16$, with Gini coefficient $G \approx 0.76$ – a highly unequal distribution, matching the qualitative content of the 80/20 rule. Rank-size examples use a power law for ordered ranks, while the Pareto density uses an exponent $\alpha + 1$; the exponent must be interpreted in the distribution where it is being used.

Where we have been

Three named functions, each the analytic extension of a discrete object, all tied together through the gamma function:

- **Gamma** $\Gamma(z)$ extends $n!$; recursion $\Gamma(z+1) = z\Gamma(z)$, reflection $\Gamma(z)\Gamma(1-z) = \pi/\sin\pi z$, Euler's limit, Stirling, digamma.
- **Beta** $B(p,q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$ is closely related to binomial coefficients and evaluates a wide class of definite integrals (including the n-ball volume $\pi^{n/2}/\Gamma(\frac{n}{2}+1)\,r^n$).
- **Zeta** $\zeta(s)$ extends the p-series; integral and Hankel-contour forms, the values $\zeta(-n) = (-1)^n B_{n+1}/(n+1)$, and the Euler product $\zeta(n) = \prod_p (1-p^{-n})^{-1}$ linking analysis to the primes.

Next: the tools of complex analysis sharpen into the study of further special functions.